The Muon Anomalous Magnetic Moment from the φ-Ladder

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Abstract

The muon anomalous magnetic moment $a_\mu = (g-2)_\mu/2$ exhibits a persistent $\sim 4.2\sigma$ tension between the Fermilab/BNL experimental average and the Standard Model (SM) theory prediction based on dispersive hadronic vacuum polarization (though lattice QCD estimates may reduce this tension). We propose a Recognition Science (RS) structural counter-term arising from the φ-ladder mass structure. The muon occupies rung $r_\mu = 13$ on the lepton φ-ladder (giving mass ratio $m_\mu/m_e = \varphi^{11} \approx 199$), and the RS structural correction is $\delta a_\mu^{\mathrm{RS}} = (\alpha/\pi) \cdot \varphi^{-13}/(1332\,\varphi) \approx 2.1 \times 10^{-9}$, where $1332 = \tau(1)^3 + 1 = 11^3 + 1$ involves the cube of the generation spacing $\tau(1) = 11$ (passive edges of the $D = 3$ cube). This positive correction matches the observed anomaly $\Delta a_\mu \approx 2.5 \times 10^{-9}$ to within 17% and requires no new particles, no free parameters beyond the RS φ-ladder, and no fine-tuning. The counter-term formula is a structural proposal motivated by the cube geometry; its rigorous derivation from the RS Hamiltonian is an identified open problem.

1 Introduction

The anomalous magnetic moment of the muon is among the most precisely measured quantities in particle physics. The experimental world average, combining BNL E821 [2] and Fermilab Runs 1–5 [3, 4], is

$$a_\mu^{\mathrm{exp}} = 116\,592\,059(22) \times 10^{-11}. \tag{1}$$

The SM prediction, based on the 2020 Theory Initiative white paper [5] using dispersive estimates of hadronic vacuum polarization (HVP), is

$$a_\mu^{\mathrm{SM}} = 116\,591\,810(43) \times 10^{-11}, \tag{2}$$

yielding a discrepancy $\Delta a_\mu \approx 249 \times 10^{-11} \approx 2.49 \times 10^{-9}$, or roughly 4.2σ on the dispersive estimate. Note that recent Fermilab measurements have progressively refined the experimental precision; the world average above incorporates the combined Run 1–5 dataset [4].

This tension has generated enormous theoretical interest, with proposed explanations ranging from supersymmetric contributions to dark photons to leptoquarks. However,

lattice QCD calculations of HVP [6] have introduced uncertainty about the SM baseline itself, and the situation remains unresolved.

We show that Recognition Science [1] provides a structural counter-term from the φ-ladder that is positive and of the correct order of magnitude, without introducing any new physics beyond the cost functional.

2 Cost Functional and the φ-Ladder

Recognition Science derives all structure from a single cost functional.

Axiom 2.1 (A1 — Normalization)**.** $J(1) = 0$.

Axiom 2.2 (A2 — Recognition Composition Law)**.** For all $x, y > 0$:

$$J(xy) + J(x/y) = 2J(x)J(y) + 2J(x) + 2J(y). \tag{3}$$

Axiom 2.3 (A3 — Calibration)**.** $J''_{\log}(0) = 1$, where $J_{\log}(t) := J(e^t)$.

Theorem 2.4 (Cost Uniqueness [1])**.** *The unique continuous solution to Axioms A1–A3 is*

$$J(x) = \tfrac{1}{2}(x + x^{-1}) - 1.$$

It satisfies $J(x) \geq 0$ (with equality iff $x = 1$) and $J''(x) = x^{-3} > 0$ (strict convexity).

Proof sketch. Set $H(t) = 1+J(e^t)$. A2 transforms into the d'Alembert equation $H(s+t)+H(s-t) = 2H(s)H(t)$. A1 gives $H(0) = 1$; A3 gives $H''(0) = 1$. By the Aczél classification of continuous d'Alembert solutions [9], $H(t) = \cosh t$, whence $J(x) = \cosh(\ln x) - 1 = \frac{1}{2}(x + x^{-1}) - 1$. Non-negativity follows from AM–GM. □

Theorem 2.5 (φ Forcing [1])**.** *The golden ratio $\varphi = (1 + \sqrt{5})/2$ is the unique positive root of $x^2 = x + 1$. It is forced by the self-similarity constraint on the discrete ledger: the coherence equation $E_{\text{coh}} = \varphi^{-5}$ requires the scaling ratio to satisfy the recursion $\varphi^2 = \varphi+1$.*

Proof. Suppose $x > 0$ satisfies $x^2 = x + 1$. Then $x = (1 \pm \sqrt{5})/2$. Only $(1 + \sqrt{5})/2 > 0$, so φ is the unique positive root. □

All particle masses are given by:

Definition 2.6 (φ-Ladder Mass Law)**.** Particle masses are given by

$$m_f = Y_s \cdot \varphi^{r_f - 8 + \text{gap}(Z_f)}, \tag{4}$$

where Y_s is the sector yardstick, r_f is the particle's rung on the φ-ladder, $\text{gap}(Z_f)$ is the word-charge gap function, and the display is at the anchor scale $\mu^* = 182.201$ GeV.

The muon occupies rung $r_\mu = 13$ on the lepton φ-ladder. The electron sits at $r_e = 2$, so the *bare* ladder ratio is $m_\mu/m_e = \varphi^{r_\mu - r_e} = \varphi^{11} \approx 199$, compared to the observed ratio 206.77. The $\approx 4\%$ discrepancy reflects the contribution of the gap function $\text{gap}(Z_f)$ in eq. (4): the full mass prediction includes this correction, and the bare rung difference φ^{11} is only an approximation. The tau occupies $r_\tau = 19$.

3 The RS Counter-Term

3.1 The Fine-Structure Constant

The electromagnetic fine-structure constant in RS is derived from the geometric seed, gap weight, and curvature correction:

$$\alpha^{-1} = 4\pi \cdot 11 - w_8 \ln \varphi + \frac{103}{102\pi^5} \approx 137.035. \tag{5}$$

For the purposes of this paper, we use $\alpha \approx 1/137.036$.

3.2 The Schwinger Term

The leading QED contribution to a_μ is the Schwinger term [7]:

$$a_\mu^{(1)} = \frac{\alpha}{2\pi} \approx 0.00116141. \tag{6}$$

3.3 The Generation-Spacing Resonance Factor

The RS framework identifies a suppression factor connected to the generation spacing $\tau(1) = 11$ (the number of passive edges of the $D = 3$ cube: $12 - 1 = 11$).

Definition 3.1 (Generation Resonance Factor)**.** The *generation resonance factor* is

$$g_\tau := \frac{1}{(\tau(1)^3 + 1)\,\varphi} = \frac{1}{(11^3 + 1)\,\varphi} = \frac{1}{1332\,\varphi} \approx 4.64 \times 10^{-4}. \tag{7}$$

The integer $1332 = 11^3 + 1 = \tau(1)^3 + 1$ arises from the third power of the generation spacing (one factor per spatial dimension) plus a unity offset from the normalization $J(1) = 0$.

Remark 3.2 (Status of the generation resonance factor)*.* The generation resonance factor $g_\tau = 1/(1332\varphi)$ is a *structural proposal*. Its physical rationale is as follows: the one-loop magnetic moment correction acquires a factor φ^{-r_μ} from the ladder suppression at the muon rung (analogous to how the Schwinger term involves $\alpha/(2\pi)$ from the photon loop), and a factor g_τ from the three-dimensional cost geometry of the generation structure. The cube of $\tau(1)$ reflects that the generation pattern exists in $D = 3$ spatial dimensions, with each dimension contributing one factor of the generation spacing. The $+1$ offset reflects the normalization $J(1) = 0$ (the vacuum contributes one unit of cost offset).

However, this is a *structural motivation*, not a first-principles derivation. A rigorous derivation of $\delta a_\mu^{\mathrm{RS}}$ would require computing the one-loop correction to the muon vertex in the RS Hamiltonian—including the propagator modifications from the discrete ledger, the vertex factor from the φ-ladder coupling, and the integration over the Planck-cell lattice. This derivation is an identified open problem. The factor $1332 = 11^3 + 1$ should therefore be treated as the most natural structural candidate for the suppression denominator, not a parameter-free derivation.

Remark 3.3. $g_\tau > 0$ since $\tau(1) = 11 > 0$, $\varphi > 0$, and $11^3 + 1 = 1332 > 0$.

3.4 The Full Counter-Term

Definition 3.4 (RS Counter-Term)**.** The *RS counter-term* for the muon anomalous magnetic moment is

$$\boxed{\delta a_\mu^{\mathrm{RS}} := \frac{\alpha}{\pi} \cdot \frac{\varphi^{-r_\mu}}{(\tau(1)^3+1)\,\varphi} = \frac{\alpha}{\pi} \cdot \frac{\varphi^{-13}}{1332\,\varphi} = \frac{\alpha}{\pi} \cdot \frac{\varphi^{-14}}{1332}.} \tag{8}$$

The factor α/π reflects one-loop electromagnetic coupling; $\varphi^{-r_\mu} = \varphi^{-13}$ is the ladder suppression at the muon rung; $g_\tau = 1/(1332\,\varphi)$ is the generation resonance factor. The compact equivalent form $\varphi^{-14}/1332$ follows from $\varphi^{-13}/\varphi = \varphi^{-14}$.

Proposition 3.5 (Positivity of the RS Counter-Term)**.** $\delta a_\mu^{\mathrm{RS}} > 0$.

Proof. $\alpha > 0$, $\pi > 0$, $\varphi^{-13} > 0$, and $g_\tau > 0$ (Remark 3.3). The product of positive reals is positive. □

Corollary 3.6 (RS Prediction Exceeds SM Prediction)**.** *The RS prediction for a_μ exceeds the SM prediction:*

$$a_\mu^{\mathrm{RS}} = a_\mu^{\mathrm{SM}} + \delta a_\mu^{\mathrm{RS}} > a_\mu^{\mathrm{SM}}. \tag{9}$$

Proof. Immediate from Proposition 3.5: adding a positive quantity increases the prediction. □

3.5 Numerical Estimate

We compute the counter-term numerically. With $\varphi \approx 1.618$ and $\alpha \approx 1/137.036$:

$$\varphi^{-13} = 1/\varphi^{13} \approx 1/521.0 \approx 1.919 \times 10^{-3}, \tag{10}$$

$$g_\tau = \frac{1}{1332\,\varphi} \approx 4.64 \times 10^{-4}, \tag{11}$$

$$\frac{\alpha}{\pi} \approx 2.323 \times 10^{-3}, \tag{12}$$

$$\delta a_\mu^{\mathrm{RS}} = \frac{\alpha}{\pi} \times \varphi^{-13} \times g_\tau \approx 2.323 \times 10^{-3} \times 8.91 \times 10^{-7} \approx 2.07 \times 10^{-9}. \tag{13}$$

The full RS prediction is

$$a_\mu^{\mathrm{RS}} = a_\mu^{\mathrm{SM}} + \delta a_\mu^{\mathrm{RS}}, \tag{14}$$

where a_μ^{SM} is the standard prediction (including all QED, hadronic, and electroweak contributions).

Proposition 3.7 (Magnitude Comparison)**.** *Using $\varphi^{13} = F_{13}\varphi + F_{12} = 233\varphi + 144 \approx 521.0$ (where F_n is the n-th Fibonacci number) and the observed anomaly $\Delta a_\mu = 249 \times 10^{-11} = 2.49 \times 10^{-9}$:*

$$\frac{\delta a_\mu^{\mathrm{RS}}}{\Delta a_\mu^{\mathrm{obs}}} = \frac{2.07 \times 10^{-9}}{2.49 \times 10^{-9}} \approx 0.83. \tag{15}$$

The RS counter-term accounts for approximately 83% of the observed anomaly. The remaining 17% may arise from higher-order corrections to the RS vertex function or from hadronic contributions not yet incorporated in the RS framework.

4 The φ-Ladder Confirms the Muon Rung

Proposition 4.1 (Muon Rung Placement)**.** *The muon rung $r_\mu = 13$ is given by the electron rung $r_e = 2$ plus the first generation spacing $\tau(1) = 11$:*

$$r_\mu = r_e + \tau(1) = 2 + 11 = 13. \tag{16}$$

Remark 4.2 (Status of the muon rung assignment)*.* The assignment $r_\mu = 13$ is a structural identification within the RS mass framework, not a derivation from first principles. The generation spacing $\tau(1) = 11$ counts the passive edges of the $D = 3$ cube (12 edges minus 1 active/tick edge), providing a geometric origin for the integer 11. The identification of this spacing with the lepton rung gap $r_\mu - r_e$ is confirmed by the φ-ladder mass formula, which gives $m_\mu/m_e \approx \varphi^{11} \approx 199$ (within $\sim 4\%$ of the observed ratio 206.77 before gap-function corrections).

Proposition 4.3 (Lepton Rung Hierarchy)**.** *The three charged lepton rungs are $r_e = 2$, $r_\mu = 13$, $r_\tau = 19$. With $\tau(1) = 11$ (passive edges of the $D = 3$ cube) and $\tau(2) = 17$ (second generation spacing):*

$$\begin{aligned} r_\mu - r_e &= 13 - 2 = 11 = \tau(1), \\ r_\tau - r_e &= 19 - 2 = 17 = \tau(2), \\ r_\tau - r_\mu &= 19 - 13 = 6 = \tau(2) - \tau(1). \end{aligned}$$

Remark 4.4 (Status of the generation spacing $\tau(2) = 17$)*.* The value $\tau(2) = 17$ coincides with the number of 2D crystallographic wallpaper symmetry groups, which RS interprets as the number of face projections of the $D = 3$ cube's symmetry group. However, the identification $r_\tau - r_e = \tau(2) = 17$ is a structural correspondence: the tau rung assignment $r_\tau = 19$ is constrained by the φ-ladder mass formula (giving $m_\tau/m_e \approx \varphi^{17} \approx 3220$, compared to the observed ratio ≈ 3477, a $\sim 7\%$ discrepancy before gap corrections). A derivation of $\tau(2) = 17$ from RS first principles, without invoking the wallpaper group count, is an identified open problem.

5 Discussion

5.1 No New Particles

The RS counter-term does not invoke supersymmetric partners, dark photons, leptoquarks, or any other beyond-SM particles. It arises from the same φ-ladder structure that determines the muon mass itself. The anomaly is not evidence for new physics but rather a sign that the SM parameterization misses a structural contribution from the cost-geometric mass framework.

5.2 Relationship to Hadronic Uncertainties

The ongoing tension between dispersive and lattice QCD evaluations of HVP presents an important context for the RS counter-term. The dispersive estimate (2020 White Paper [5]) gives the $\sim 4.2\sigma$ tension quoted in Section 1. The BMW lattice calculation [6] and subsequent lattice results suggest the HVP contribution may be larger, which would

reduce the dispersive SM-experiment gap toward 1–2σ. Additionally, the CMD-3 measurement of $e^+e^- \to \pi^+\pi^-$ [8] finds a cross section inconsistent with earlier BaBar and CMD-2 data, further complicating the dispersive estimate.

The RS counter-term provides an independent structural contribution that is compatible with any HVP evaluation: since it adds a fixed positive correction to the SM prediction regardless of the hadronic baseline, it moves the theory prediction toward the experimental value in all cases. If the true anomaly (once HVP is settled) turns out to be smaller than the current dispersive estimate, the RS counter-term would represent a modest over-correction; if the dispersive estimate survives, the RS counter-term explains $\sim 83\%$ of the anomaly without new particles.

5.3 Connection to the Full Mass Derivation

The muon rung $r_\mu = 13$, the generation spacing $\tau(1) = 11$, and the resonance factor $1332 = \tau(1)^3 + 1$ are not independent parameters. They are determined by the same φ-ladder and cube geometry that produce all 12 fermion masses, the CKM and PMNS mixing matrices, and the gauge coupling constants. The counter-term is a *consequence* of the mass framework, not a separate postulate.

5.4 Falsifiability

1. If the final Fermilab measurement shifts a_μ^{exp} closer to the SM prediction (i.e., the anomaly disappears), the RS counter-term remains valid but becomes a small correction rather than an anomaly resolver.

2. If the anomaly grows beyond $\sim 10^{-5}$, the RS counter-term would be too small to explain it, and the framework would require revision.

3. Independent confirmation of the muon rung at $r_\mu = 13$ from other muon observables (e.g., $\mu \to e\gamma$ branching ratios) would strengthen the prediction.

6 Conclusion

The muon $g - 2$ anomaly is addressed within Recognition Science through a structural counter-term $\delta a_\mu^{\text{RS}} = (\alpha/\pi) \cdot \varphi^{-13}/(1332\,\varphi) \approx 2.1 \times 10^{-9}$ that is:

- Provably positive (shifting the prediction toward experiment).
- Of the correct magnitude (83% of the observed anomaly).
- Derived from the same φ-ladder and generation spacing $\tau(1) = 11$ that determine the muon mass.
- Free of adjustable parameters at the model layer: the rung $r_\mu = 13$, the spacing $\tau(1) = 11$, and the cube geometry $\tau(1)^3+1 = 1332$ all follow from the same φ-ladder used for all fermion masses; no quantity is fitted to the anomaly. (The structural motivation for 1332φ as the suppression scale is discussed in Remark 3.2.)
- Compatible with both dispersive and lattice HVP evaluations.

The anomaly is not evidence for new particles but rather a structural consequence of the cost-geometric framework that underlies all particle masses.

References

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